

大数定律、中心极限定理和集中不等式

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1. (1) 令随机变量 $X = \sum_{i=1}^{30} X_i, X_i \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(3), i = 1, 2, \dots, 30$. 易知 $X \sim \Gamma(30, 3), \mathbb{E}(X) = 10, \text{Var}(X) = 30\text{Var}(X_i) = \frac{10}{3}$.

下面利用中心极限定理估计概率.

$$\begin{aligned} P(9 < X \leq 11) &\approx \Phi(\sqrt{0.3}) - \Phi(-\sqrt{0.3}) \\ &= 2\Phi(\sqrt{0.3}) - 1 \\ &\approx 0.4161 \end{aligned}$$

- (2) 由中心极限定理:

$$\frac{X - \frac{1}{3}n}{\frac{1}{3}\sqrt{n}} \xrightarrow{d} N(0, 1)$$

因此 $X \approx Z \sim N(\frac{n}{3}, \frac{n}{9})$.

$$\begin{aligned} P(X \leq 12) &\approx \Phi\left(\frac{12 - \frac{n}{3}}{\frac{1}{3}\sqrt{n}}\right) \\ &= \Phi\left(\frac{36}{\sqrt{n}} - \sqrt{n}\right) \end{aligned}$$

令 $\Phi(\frac{36}{\sqrt{n}} - \sqrt{n}) \geq 0.95$ 得 $n \leq 27$, 即 n 最大可为 27.

2. (1) 由题可知:

$$\forall \epsilon > 0, \lim_{n \rightarrow +\infty} P\left(\bigcup_{m=n}^{+\infty} A_m(\epsilon)\right) = 0, A_m(\epsilon) = \left\{\left|\frac{\ln X_m}{m} - c\right| \geq \epsilon\right\}$$

由 Union Bound 及 Markov 不等式可得:

$$\begin{aligned} P\left(\bigcup_{m=n}^{+\infty} A_m(\epsilon)\right) &\leq \sum_{m=n}^{+\infty} P(A_m(\epsilon)) \\ &\leq \sum_{m=n}^{+\infty} \frac{\mathbb{E}(\ln X_m) - mc}{m\epsilon} \end{aligned}$$

由题可得: $\mathbb{E}(\ln X_0) = 0$,

$$\begin{aligned} \mathbb{E}(\ln X_m) &= p\mathbb{E}(\ln(aX_{m-1})) + (1-p)\mathbb{E}(\ln(bX_{m-1})) \\ &= p\ln a + (1-p)\ln b + \mathbb{E}(\ln X_{m-1}) \end{aligned}$$

因此, $\mathbb{E}(\ln X_m) = (p\ln a + (1-p)\ln b)m$.

综上可得, 对 $\forall \epsilon > 0$,

$$\lim_{n \rightarrow +\infty} P\left(\bigcup_{m=n}^{+\infty} A_m(\epsilon)\right) \leq \lim_{n \rightarrow +\infty} \sum_{m=n}^{+\infty} \frac{p \ln a + (1-p) \ln b - c}{\epsilon}$$

故当 $c = p \ln a + (1-p) \ln b$ 时 $\ln X_n$ 几乎必然收敛到 c .

(2) 由题可知: $X_m = \prod_{i=1}^m A_i, A_i = \begin{cases} a, & p \\ b, & 1-p \end{cases}$ 且 A_i 相互独立.

故随机变量 $Y_m = \ln X_m = \sum_{i=1}^m \ln A_i$.

由中心极限定理:

$$Y_{913} = \sum_{i=1}^{913} \ln A_i \approx Y \sim N(n\mu, n\sigma^2)$$

其中,

$$\mu = \mathbb{E}(\ln A_i) = p \ln a + (1-p) \ln b = 0.5 \ln 0.99 \approx -0.005025$$

$$\sigma^2 = \text{Var}(\ln A_i) = \mathbb{E}((\ln A_i)^2) - \mu^2 \approx 0.010067$$

$$\text{故 } P(X_{913} \geq 1) = P(Y_{913} \geq 0) \approx 1 - \Phi\left(\frac{-913\mu}{\sqrt{913\sigma^2}}\right) \approx 0.065103$$

3. 由题可知, 对 $\forall \epsilon > 0$,

$$\lim_{n \rightarrow +\infty} P\left(\left|\frac{1}{2n} \sum_{i=1}^{2n} X_i - c\right| \geq \epsilon\right) = 0$$

令 $X_1 = \frac{1}{2n} \sum_{k=1}^n X_{2k-1}, X_2 = \frac{1}{2n} \sum_{k=1}^n X_{2k}$, 则 $X = \frac{1}{2n} \sum_{i=1}^{2n} X_i = X_1 + X_2$.

易得 $\mathbb{E}(X_1) = 0.6, \text{Var}(X_1) = \frac{0.18}{n}, \mathbb{E}(X_2) = 0.5, \text{Var}(X_2) = \frac{1}{3n}$.

由 Chebyshev 不等式, 对 $\forall \epsilon > 0$,

$$\begin{aligned} P(|X_1 - 0.6| \geq \epsilon) &\leq \frac{0.18}{n\epsilon^2} \\ P(|X_2 - 0.5| \geq \epsilon) &\leq \frac{1}{3n\epsilon^2} \end{aligned}$$

另一方面, 由三角不等式及 X_1, X_2 独立可得: 对 $\forall \epsilon > 0$,

$$\begin{aligned} P(|X - c| \geq \epsilon) &= P(|X_1 - c_1 + X_2 - c_2| \geq \epsilon) \\ &\leq P(|X_1 - c_1| + |X_2 - c_2| \geq \epsilon) \\ &\leq P\left(|X_1 - c_1| \geq \frac{\epsilon}{2}\right) + P\left(|X_2 - c_2| \geq \frac{\epsilon}{2}\right) \end{aligned}$$

其中 $c = c_1 + c_2$.

注意到当 $c_1 = 0.6, c_2 = 0.5$ 时, 对 $\forall \epsilon > 0, P(|X - c| \geq \epsilon) \rightarrow 0, (n \rightarrow +\infty)$.

此时 $c = c_1 + c_2 = 1.1$.

4. (1) 只需证对 $\forall \epsilon > 0, P(|F_n(x) - F(x)| \geq \epsilon) \rightarrow 0, (n \rightarrow +\infty)$.

对 $\forall x \in \mathbb{R}, F_n(x)$ 为在 $[0, 1]$ 上取值的离散型随机变量.

注意到 $\mathbb{E}(F_n(x)) = \mathbb{E}(1_{\{X_1 \leq x\}}) = F(x)$.

因此由 Chebyshev 不等式: 对 $\forall \epsilon > 0$,

$$P(|F_n(x) - F(x)| \geq \epsilon) \leq \frac{\text{Var}(F_n(X))}{\epsilon^2} = \frac{\text{Var}(1_{\{X_1 \leq x\}})}{n\epsilon^2} \rightarrow 0, (n \rightarrow +\infty)$$

□

(2) 令随机变量 $X = \frac{1}{n} \sum_{i=1}^n f(X_i)$.

注意到 $\mathbb{E}(X) = \mathbb{E}(f(X_1)) = \int_0^1 f(x)dx$.

由 $f(x)$ 在 $[0, 1]$ 上连续知 $\text{Var}(f(X_1))$ 存在, 故由 Chebyshev 不等式: 对 $\forall \epsilon > 0$,

$$P(|X - \int_0^1 f(x)dx| \geq \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2} = \frac{\text{Var}(f(X_1))}{n\epsilon^2} \rightarrow 0, (n \rightarrow +\infty)$$

□

5. (1) 设随机变量 $X_i \sim B(1, \mu_1)$, 则 $N_1 = \sum_{i=1}^m X_i$, $\mathbb{E}(N_1) = m\mu_1$. 由 Hoeffding 不等式:

$$\begin{aligned} P(|\mu_1 - \hat{\mu}_1| \geq \epsilon) &= P(|N_1 - \mathbb{E}(N_1)| \geq m\epsilon) \\ &\leq P(N_1 - \mathbb{E}(N_1) \geq m\epsilon) + P(N_1 - \mathbb{E}(N_1) \leq -m\epsilon) \\ &\leq 2e^{-\frac{2m^2\epsilon^2}{m}} = 2e^{-2m\epsilon^2} \end{aligned}$$

□

(2) 设随机变量 $Y_i \sim B(1, \mu_2)$, 则 $N_2 = \sum_{i=1}^m Y_i$, $\mathbb{E}(N_2) = m\mu_2$. 由题可知:

$$\begin{aligned} P(A) &= 1 - P((\mu_1 - \mu_2)(\hat{\mu}_1 - \hat{\mu}_2) < 0) \\ &= 1 - P((\mu_1 - \mu_2)(N_1 - N_2) < 0) \end{aligned}$$

因此只需证 $P((\mu_1 - \mu_2)(N_1 - N_2) < 0) \leq e^{-m\Delta^2}$.

注意到 $\mathbb{E}((\mu_1 - \mu_2)(N_1 - N_2))$ 恰好等于 $m\Delta^2$. 由 Hoeffding 不等式:

$$\begin{aligned} P((\mu_1 - \mu_2)(N_1 - N_2) < 0) &= P((\mu_1 - \mu_2)(N_1 - N_2) - m\Delta^2 < -m\Delta^2) \\ &\leq e^{-\frac{2m^2\Delta^4}{m(2\Delta^2)}} = e^{-m\Delta^2} \end{aligned}$$

□

6. (1) 由题可知:

$$\begin{aligned} Z_n \xrightarrow{\text{a.s.}} Z &\iff P(\lim_{n \rightarrow +\infty} Z_n = Z) = 1 \\ &\iff P(\forall \omega \in S, \lim_{n \rightarrow +\infty} Z_n(\omega) = Z(\omega)) = 1 \end{aligned}$$

由随机变量的定义可知: $\{Z_n(\omega)\}_{n=1}^{+\infty} \in \mathbb{R}$ 为实数列, $Z(\omega) \in \mathbb{R}$.

故由柯西收敛准则:

$$\begin{aligned} \forall \omega \in S, \lim_{n \rightarrow +\infty} Z_n(\omega) &= Z(\omega) \\ \iff \forall \omega \in S, \epsilon > 0, \exists N, \text{s.t.} \forall n, m > N, |Z_n(\omega) - Z_m(\omega)| &< \epsilon \\ \iff \forall \epsilon > 0, \exists N, \text{s.t.} \forall n, m > N, |Z_n - Z_m| &< \epsilon \end{aligned}$$

因此, 根据事件几乎必然成立的定义:

$$Z_n \xrightarrow{\text{a.s.}} Z \iff \forall \epsilon > 0, \exists N, \text{s.t.} \forall n, m > N, |Z_n - Z_m| < \epsilon \text{ 几乎必然成立.} \quad \square$$

(2) 根据 $P(X_1 > 0) = 1$ 以及 X_i 独立同分布可知 S_n 关于 n 严格单调递增.

又由 $Y_t = \max_{S_n \leq t} n$ 知 Y_t 同样关于 t 单调递增且

$$S_{Y_t} \leq t \leq S_{Y_{t+1}}$$

故

$$\frac{S_{Y_t}}{Y_t} \leq \frac{t}{Y_t} \leq \frac{S_{Y_{t+1}}}{Y_{t+1}} \cdot \frac{Y_{t+1}}{Y_t}$$

两边同时取极限得:

$$\mu = \lim_{t \rightarrow +\infty} \frac{S_{Y_t}}{Y_t} \leq \lim_{t \rightarrow +\infty} \frac{t}{Y_t} \leq \lim_{t \rightarrow +\infty} \frac{S_{Y_{t+1}}}{Y_{t+1}} \cdot \frac{Y_{t+1}}{Y_t} = \mu$$

因此有

$$\lim_{t \rightarrow +\infty} \frac{Y_t}{t} = \frac{1}{\mu}$$

□

7. 定义经验分布函数

$$F_n(x) = \frac{1}{2n+1} \sum_{i=1}^{2n+1} 1_{\{X_i \leq x\}}$$

由于 X_i 独立同分布, 故对于 $\forall x, 1_{\{X_i \leq x\}}$ 独立同分布, 由强大数定律:

$$F_n(x) \xrightarrow{\text{a.s.}} \mathbb{E}(1_{\{X_i \leq x\}}) = F(x)$$

因此有

$$P\left(\lim_{n \rightarrow +\infty} F_n(M_n) = F(M_n)\right) = 1$$

$$P\left(\lim_{n \rightarrow +\infty} F_n(c) = F(c)\right) = 1$$

注意到 $F(c) = 0.5$ 且 $\lim_{n \rightarrow +\infty} F_n(M_n) = \lim_{n \rightarrow +\infty} \frac{n+1}{2n+1} = 0.5$.

又因为 $\exists! c \in \mathbb{R}, \text{s.t.} F(c) = 0.5$, 故 $P(\lim_{n \rightarrow +\infty} M_n = c) = 1$. □

8. (1) 由 $X_n \xrightarrow{P} X$ 知:

$$\forall \epsilon > 0, \lim_{n \rightarrow +\infty} P(A_n(\epsilon)) = 0, A_n(\epsilon) = \{|X_n - X| \geq \epsilon\}$$

令 $a_n = P(A_n(\epsilon))$.

由于 $\{a_n\}$ 收敛到 0, 因此 $\exists \{X_{n_k}\}_{k=1}^{+\infty} \subseteq \{X_n\}_{n=1}^{+\infty}, \text{s.t.} \sum_{k=1}^{+\infty} P(A_{n_k}(\epsilon))$ 有界 (例如取一组足够大的 n_k 满足 $P(A_{n_k}(\epsilon)) < 2^{-k}$).

由 Borel-Cantelli 引理:

$$P\left(\bigcap_{m=1}^{+\infty} \bigcup_{k=m}^{+\infty} A_{n_k}(\epsilon)\right) = 0$$

由此通过反证法可得：

$$\forall \epsilon > 0, \lim_{m \rightarrow +\infty} P\left(\bigcup_{k=m}^{+\infty} A_{n_k}(\epsilon)\right) = 0$$

也即 $P(X_{n_k} \xrightarrow{\text{a.s.}} X) = 1$. □

(2) $\Rightarrow$: 根据依概率收敛的定义：

$$X_n \xrightarrow{P} X \iff \forall \epsilon > 0, \lim_{n \rightarrow +\infty} P(|X_n - X| \geq \epsilon) = 0$$

由三角不等式及 Union Bound 可得：

$$\begin{aligned} \lim_{n, m \rightarrow +\infty} P(|X_n - X_m| \geq \epsilon) &\leq \lim_{n, m \rightarrow +\infty} P(|X_n - X| + |X_m - X| \geq \epsilon) \\ &\leq \lim_{n \rightarrow +\infty} P(|X_n - X| \geq \frac{\epsilon}{2}) + \lim_{m \rightarrow +\infty} P(|X_m - X| \geq \frac{\epsilon}{2}) \\ &= 0 \end{aligned}$$

$\Leftarrow$: 由条件可得： $\forall \epsilon, \delta > 0, \exists N, \text{s.t. } \forall n, m > N, P(|X_n - X_m| \leq \epsilon) > 1 - \delta$.

即当 $n, m \rightarrow +\infty$ 时， $|X_n - X_m| \leq \epsilon$ 几乎必然成立.

取 $m = N + 1, \epsilon = 1$ 得：

$$\begin{aligned} |X_n - X_{N+1}| \leq 1 &\implies |X_n| - |X_{N+1}| \leq 1 \\ &\implies |X_n| \leq 1 + |X_{N+1}| \\ &\iff \{X_n\} \text{ 有界} \end{aligned}$$

由上可知： $\{X_n\}$ 几乎必然有界. 由有界实数列必存在收敛子列知：

几乎必然存在收敛子列 $\{X_{n_k}\}_{k=1}^{+\infty} \subseteq \{X_n\}_{n=1}^{+\infty}$.

设该子列收敛到 X ，即当 $\forall \epsilon > 0, \exists K, \text{s.t. } \forall k > K, |X_{n_k} - X| < \frac{\epsilon}{2}, |X_n - X_{n_k}| < \frac{\epsilon}{2}$.

根据三角不等式即可得 $\forall \epsilon > 0, \exists N = n_K, \text{s.t. } \forall n > N, |X_n - X| < \epsilon$. 即 X_n 收敛到 X .

综上，当 $n \rightarrow +\infty$ 时， X_n 收敛到 X 几乎必然发生，即 $\exists X, \text{s.t. } X_n \xrightarrow{P} X$. □

9. (1) 由 Markov 不等式：

$$P\left(\sum_{n=1}^m X_n > A\right) \leq \frac{\mathbb{E}(\sum_{n=1}^m X_n)}{A}$$

由题可知： $\mathbb{E}(\sum_{n=1}^m X_n)$ 单调递增且 $\lim_{m \rightarrow +\infty} \mathbb{E}(\sum_{n=1}^m X_n) \leq M$. 故

$$\forall m \geq 1, P\left(\sum_{n=1}^m X_n > A\right) \leq \frac{M}{A}$$

$$\lim_{A \rightarrow +\infty} P\left(\sum_{n=1}^{\infty} X_n > A\right) = 0$$

即

$$\forall \epsilon > 0, \exists A_0, \text{s.t. } \forall m \geq 1, A > A_0, P\left(\sum_{n=1}^m X_n > A\right) < \epsilon$$

综上所述， $P(\forall A > 0, \exists m, \text{s.t. } \sum_{n=1}^m X_n > A) = 0$.

即 $P(\sum_{n=1}^{+\infty} X_n < +\infty) = 1$. □

(2) 令 $S_m = \sum_{n=1}^m X_n, S = \lim_{m \rightarrow +\infty} S_m$.

由题可知: $\mathbb{E}(S_m) = \sum_{n=1}^m \mathbb{E}(X_n), \mathbb{E}(S)$ 发散, $\text{Var}(S_m) = \sum_{n=1}^m \text{Var}(X_n)$.

由 Chebyshev 不等式:

$$P(\mathbb{E}(S_m) - S_m \geq \delta \mathbb{E}(S_m)) \leq P(|S_m - \mathbb{E}(S_m)| \geq \delta \mathbb{E}(S_m)) \leq \frac{\text{Var}(S_m)}{\delta^2 \mathbb{E}^2(S_m)}$$

其中 $\delta \in (0, 1)$ 为常数. 由题可知:

$$\lim_{m \rightarrow +\infty} P(S_m \leq (1 - \delta) \mathbb{E}(S_m)) = 0$$

故必然存在子列 $\{S_{n_k}\}$ 使得

$$\sum_{k=1}^{+\infty} P(S_{n_k} \leq (1 - \delta) \mathbb{E}(S_{n_k})) \leq M$$

M 为常数. 根据 Borel-Cantelli 引理:

$$\lim_{k \rightarrow +\infty} P(S_{n_k} \leq (1 - \delta) \mathbb{E}(S_{n_k}) \rightarrow +\infty) = 0$$

由于 S_n 关于 n 单调递增, 因此有

$$P(S = +\infty) = \lim_{k \rightarrow +\infty} P(S_{n_k} > (1 - \delta) \mathbb{E}(S_{n_k}) \rightarrow +\infty) = 1$$

□

(3) 由题可知: X_i 两两独立且 $\mathbb{E}(X_i) = \frac{1}{2i}, \text{Var}(X_i) = \frac{1}{12i^2}$. 故

$$\sum_{i=1}^{+\infty} \mathbb{E}(X_i) = +\infty$$

$$\lim_{n \rightarrow +\infty} \frac{\sum_{i=1}^n \text{Var}(X_i)}{(\sum_{i=1}^n \mathbb{E}(X_i))^2} = \frac{1}{3} \lim_{n \rightarrow +\infty} \frac{\sum_{i=1}^n \frac{1}{i^2}}{(\sum_{i=1}^n \frac{1}{i})^2} = 0$$

由 (2) 可知: $\sum_{i=1}^{+\infty} X_i$ 几乎必然发散到 $+\infty$.

(4) 不能, 因为只有事件相互独立才能将尾事件概率分解为单个事件概率乘积.